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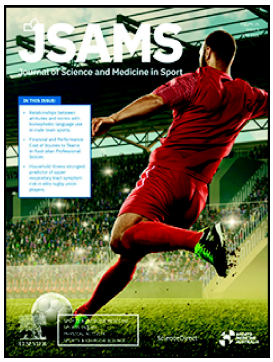

The Impact of Body Fat on Thermoregulation During Exercise in the Heat: A Systematic Review and Meta-analysis

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Title: The Impact of Body Fat on Thermoregulation During Exercise in the Heat: A Systematic Review and Meta-analysis

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ABSTRACT

PURPOSE: we performed a systematic review and meta-analysis to examine the effect of body fat on thermoregulatory responses to exercise in the heat.

METHODS: We systematically searched databases (N=8) for studies that included: healthy participants; an aerobic exercise intervention under heat stress; a measure of body fat by a high (Hi-BF) and low (Lo-BF) body fat group; pre- and post-exercise internal temperature (Tcore), and/or post-exercise mean skin temperature (Tsk) and whole body sweat loss (WBSL). Criteria to distinguish Hi-BF and Lo-BF were determined by each qualifying study. Out of 1,222 potential studies, 10 qualified. Robust variance estimation models were utilized to determine the pooled effect size (ES, 95% C.I) for interventions that performed unbias comparisons between groups.

RESULTS: Participants (n=211) were mostly male (70%) and young (19.3±7.1years) with a maximal oxygen consumption of 50.42 ± 9.3 $ml \cdot kg^{-1} \cdot min^{-1}$ by Hi-BF (36.7±11.8%) and Lo-BF (17.8±5.7%). Exercise duration was between 30-60min at moderate-high intensity (44.1-71.4%VO_2max) at 28-40.3°C and 26-50% humidity. Of the 7 interventions that normalized metabolic heat production to body mass and 5 interventions that normalized to external workload, there were no differences in any of the three outcome variables between groups.

CONCLUSION: Many of the included studies did not control for confounding factors that may affect the relationship between body fat and thermoregulation during exercise in the heat such as metabolic heat production and body mass. Given the small number of interventions included, future work must make unbiased comparisons to appropriately assess the independent influence of body fat.

Key Words: Adiposity, Body Fat, Heat Stress, Physical Activity

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1. Introduction

Since 1999 the prevalence of obesity in U.S has increased 2-4% annually and now totals an alarming 42.4% of adults [1]. Obesity gives rise to over 200 chronic diseases and health conditions that include the top causes of mortality in the U.S. including cancer, cardiovascular disease, and diabetes mellitus [2,3]. Another comorbidity of obesity that has recently come to light due to our rapidly changing climate is thermoregulatory dysfunction, or the inability to regulate internal body temperature appropriately [4]. For example, Bedno et al. (2014) reported U.S. Army recruits who exceeded U.S. Army body fat standards were 3.98 (95% CI:2.17, 7.29) times more likely to develop exertional heat illness (EHI) than normal weight recruits [5]. Several exertional heat illness (EHI) position statements and epidemiological studies now classify obesity and overweight as risk factors for EHI, however, the evidence surrounding this classification is limited and inconclusive [5–9].

The etiology of increased body fat as a risk factor for EHI stems from differences in specific heat capacity and conductivity of fat versus fat-free tissue [10]. Core temperature responses to a given heat load are strongly influenced by body mass, which serves an internal "heat sink". Further, fat-free tissue or "lean" mass has a greater specific heat capacity than fat tissue. A greater specific heat capacity within a given tissue increases its ability to transfer heat to the body surface [10]. Therefore, individuals with higher body fat would theoretically have a reduced capacity to transfer heat resulting in increased thermal strain during exercise in the heat [11]. Impaired ability to transfer heat can negatively affect thermoregulatory effector responses such as mean skin temperature (Tsk), core temperature (Tcore) and whole body sweat loss (WBSL). Impaired thermoregulatory responses to a given thermal stimulus may increase vulnerability to hyperthermia and potentially EHI [12].

It is important to note that the etiology of adiposity as a risk factor for thermoregulatory dysfunction is theoretically based and the literature surrounding body fat as an independent predictor of thermoregulatory responses has not been systematically examined. Laboratory investigations have examined the effect of body fat on thermoregulation during exercise in the heat, but these studies are

limited in number with small sample sizes with conflicting conclusions [5,9,13,14]. Some studies reported significant differences in Tcore and Tsk between high-body fat (Hi-BF) and low-body fat (Lo-BF) [15–17]where others suggest there are no differences in any thermoregulatory outcome[18–21]. Moreover, individuals with higher body fat tend to have lower fitness levels, greater body mass and a smaller body surface area to mass ratio compared to those with lower body fat, which makes it difficult to ascertain the independent influence of body fat on thermoregulatory responses during exercise in the heat[12].

To examine the effect of subject characteristics (i.e. body fat) among research participants, Cramer & Jay (2015) proposed that the experimental approach is critical to control for individual variability in temperature regulation and responses to exercise-induced heat stress[22]. In a study to examine the individual variability in thermoregulatory responses, Cramer & Jay (2015) reported that biophysical parameters such as total body mass explain 54-71% of the individual variability in Tcore and WBSL [22]. This would suggest that isolating the independent influence of body fat on thermoregulatory responses requires an aerobic exercise intervention that accounts for differences in biophysical factors between groups. Therefore, conflicting results regarding the influence of body fat on thermoregulation during exercise in the heat could be primarily attributed to study design and the absence of adjustments for biophysical factors.

The purpose of this high-quality systematic review and meta-analysis is to better ascertain the independent influence of body fat on thermoregulation during exercise in the heat in healthy populations. We aim to critically assess the literature through a meta-analysis and corresponding narrative synthesis. To independently assess the influence of body fat, a meta-analysis was performed for interventions that normalized metabolic heat production to total body mass to compare Tcore and Tsk responses between groups. Similarly, a meta-analysis of interventions that normalized metabolic heat production to external workload was performed to assess WBSL between groups. These study designs allow unbias comparisons between groups. Moreover, a systematic evaluation of the current literature and corresponding meta-analyses will assist future investigations on addressing gaps in knowledge and potential confounding variables that affect the ability to *independently* assess the effect of body fat on thermoregulation in the

heat. We hypothesized that Hi-BF individuals will experience greater thermal strain (i.e. higher Tcore, Tsk, and WBSL) than Lo-BF individuals. Additionally, thermoregulatory responses between groups will differ based on the study design and exercise prescription.

Methods

This systematic review is consistent with the *P*referred *R*eporting *I*tems for *S*ystematic *R*eviews and *M*eta-*A*nalyses (*PRISMA*) statement (11,12). We performed a Boolean search with search terms related to obesity, thermoregulation, heat stress, and exercise (Appendix A). Articles published in English were searched in PubMed, SCOPUS, SportDiscus. Cumulated Index to Nursing and Allied Health Literature (CINAHL), Academic Search Premier, and the Cochrane Library. Articles published in Spanish were searched in Scientific Electronic Library Online (SciELO) and Latin American and Carribbean Health Science. The searches were performed with the help of a medical librarian (J.L), and were restricted to publications dates between 1980 to May 23rd, 2021 to represent the current literature in this area of investigation.

Studies were peer reviewed and published in English or Spanish. Studies were eligible if they included: (1) human participants (aged 7 or over) without chronic diseases and health conditions; (2) an aerobic exercise intervention under heat stress (ambient temperature ≥ 25.0°C and ≥30% relative humidity); (3) a Hi-BF and Lo-BF group; (4) a valid assessment of body fat percentage [23,24] (DEXA, hydrostatic weighing); and (5) pre and -post exercise core temperature (ΔTcore), post-exercise mean skin temperature (Tsk) or whole body sweat loss (WBSL). Studies were excluded if they: (1) reported use of pharmacological agents that influence body temperature such as amphetamines, antihypertensives, anticholinergics, acetaminophen, diuretics, NSAIDs, or aspirin; (2) involved populations with a known sweating disorder; or (3) included exercise modalities other than aerobic exercise. Potential articles were screened first by title and abstract and then by full text. Two independent coders triaged the records. Reference lists of included studies were manually searched for additional articles.

Data were extracted using a standardized coding form[25]. We coded characteristics related to study design, participants description, % body fat, exercise intensity, exercise duration, environmental

conditions, and thermoregulatory outcomes (Tcore, Tsk, and WBSL). When studies had multiple experimental arms, they were treated as independent interventions. Each aerobic exercise intervention was characterized based on exercise intensity prescription. There are numerous methods for prescribing exercise intensity to examine thermoregulatory responses during exercise. Exercise intensity prescriptions using absolute or relative (to total body mass, surface area, lean body mass) metabolic heat production are common methods to control for individual variability between subjects. Metabolic heat production is measured through direct or indirect calorimetry[26]. Many practical exercise prescription methods such as absolute workload and relative maximal oxygen consumption or peak (VO2max or VO2peak) are also used in thermal physiology literature (Supplemental Table 1).

We contacted authors if the data necessary for calculating effect sizes (ES) of the primary outcomes were only reported in figure format. For studies whose authors did not respond to contact requests or that were unable to provide data, we obtained the data from figures using a validated software WebPlot Digitizer (Version 4.2, Oakland, CA) [27].

We evaluated the methodological quality of each study using the modified Downs and Black checklist [28] as the Cochrane risk of bias tools [29] were not applicable due to the design of the included studies. Questions within the Downs and Black checklist were modified to reflect the objectives and thermoregulatory outcome variables evaluated within the systematic review. The Downs and Black checklist was modified to fit the objective of the systematic review. The Downs and Black checklist categories of possible bias include: 1) reporting; 2) exertional validity; 3) bias (internal validity); 4) confounding; and 5) power. Methodological study quality was reported as percentage of items satisfied out of a possible 31 items. The overall methodological quality was classified as: low (≤15 points,<50%),moderate (>15−24 points, 50−79%), and high (≥25 points,≥80%)[30]. A detailed description of each of the modified Downs and Black checklist items is provided elsewhere [28]. Two coders (M.C.M., E.F.Z.) independently extracted and entered study information. All disagreements were resolved through discussion.

2.1 Statistical Analyses

Descriptive statistics are reported as weighted mean ± standard deviation (SD). To account for effect size (ES) dependencies (i.e. multiple interventions in one study), a robust variance estimation (RVE) model was utilized to determine the pooled ES for ΔTcore, Tsk, and WBSL between Hi-BF and Lo-BF using STATA software (version 15, StataCorp, College Station, USA) [31]. Only studies that allowed for unbias comparisons were included in the RVE model. SD were reported as the pooled SD of change between groups for Tcore and WBSL and absolute SD for Tsk. The absolute magnitude of the ES was interpreted as: <0.20, trivial; 0.20–0.49, small; 0.50–0.79, moderate; and ≥0.80, large. Individual study ES were calculated using hedge's g. For all significant ES factors such as age, fitness, and relative intensity of the exercise intervention were entered separately as moderators to determine whether these factors may have influenced the thermoregulatory response to exercise. Significance was set *a priori* at $p<0.05$.

2. Results

Of the 1,222 potentially studies that emerged from our search , 10 qualified (Figure 1) [14–21,32,33]. Five studies had multiple exercise intensity intervention arms, which yielded 18 interventions [15,18,21,32]. Descriptive characteristics and thermoregulatory outcomes are presented in Table 1. There was a total of 211 participants (mean= 19 ± 6 participants per study, range= 5-34 participants per study) who were mostly male (70%), unacclimatized to the heat, and physically inactive (70%). Six of the studies included adults (mean age: 25± 1 years) and four included pubescent (n=2) or prepubescent (n=2) children (mean age:12 ± 2 years). The overall % body fat (BF%) for the Hi-BF group (n=107) was 36.7 ± 11.8% (range: 22-68.8%) and 17.69 ± 5.74% (range: 10.8-26.3%) for the Lo-BF group (n= 104), with a difference between the groups of 19.01± 8.8% ($p<0.001$). VO_2max was 50.42 ± 9.3 $ml \cdot kg^{-1} \cdot min^{-1}$ which was poor to excellent for men of this age and good to superior for women[34].

Of the 18 interventions, approximately 16% used absolute metabolic heat production (k=3), 6% absolute workload (k=1), 38% metabolic heat production relative to total body mass (k=7), 6% fixed metabolic heat production relative to lean body mass (k=1), 6% fixed metabolic heat production relative to body surface area (k=1), and 28% used relative VO_2max or VO_2peak (k=5) as the measure of exercise intensity (Figure 2). Post exercise Tcore reported in the included studies did not exceed 38°C (37.72 ± 0.38°C). Environmental temperature and humidity ranged from 25-40.3°C and 26-50%, respectively. All exercise protocols were between 30-60 minutes with an overall relative exercise intensity between 44.1 and 71.4% VO_2max.

Using the Downs and Black checklist assessment tool, we found that eight studies exhibited moderate methodological quality[14–19,21,32] and two studies exhibited low methodological quality [20,33] (Supplemental Table 2). Methodological quality ranged from 45-61% of the items satisfied. None of the 10 studies were characterized as randomized controlled trials.

RVE revealed no differences in ΔTcore (ES=0.21, 95% C.I. -0.53-0.96, p=0.370), Tsk, (ES=0.799, 95% C.I. -3.06-4.66, p=0.231), or WBSL (ES=0.28, C.I. -1.093-1.67, p=0.46) between groups in response to an aerobic exercise intervention in the heat. Only 7 out of 18 interventions normalized metabolic heat production to total body mass and only 3 out of 18 interventions normalized exercise intensity to external workload.

3. Discussion

The evidence surrounding the effect of body fat on thermoregulation during exercise in the heat is limited; the purpose of this systematic review and meta-analyses was to overcome this limitation and characterize factors that may influence the relationship between body fat and thermoregulatory outcomes during exercise in the heat in a healthy population. Overall, the included studies were heterogeneous. We must recognize the study designs or subject characteristics in the current literature confound our understanding of the independent influence of body fat on thermoregulatory outcomes. Understanding these factors not only benefits the development of future studies, but it also sheds light on the whether or not body fat can

independently predict thermoregulatory outcomes and continue to be characterized as a risk factor for impaired thermoregulation in the heat. Interestingly, RVE revealed that there were no differences in ΔTcore, Tsk, and WBSL between Hi-BF and Lo-BF groups in interventions that performed unbias comparisons between Hi-BF and Lo-BF groups.

Prescribed exercise intensities between interventions of included studies were extremely diverse (Figure 2), which could have contributed to the inter-study and potentially inter-individual variability of thermoregulatory outcomes between groups. Higher variability among data reduces the ability to detect statistical significance [35]. Traditionally, researchers in environmental physiology prescribed exercise based on the participant's relative VO_2max to perform unbias comparisons between groups [36–38]. This method was implemented to reduce bias between participants of different fitness levels[36]. Although this method may provide valuable information in settings that require self-paced work[12], the approach is worrisome as it often results in differences in requirements for heat loss [36]. Jay & Cramer (2015) proposed an optimal between group study design to control for factors that modulate thermoregulatory responses during exercise in the heat by utilizing metabolic heat production. Fixing metabolic heat production among participants suggests that each participant is experiencing the same thermal stimulus and differences in heat loss can be attributed to the independent factor of interest [36]. Although 12 of the 18 interventions in this systematic review prescribed exercise intensity based on metabolic heat production [14–16,32], only 7 accounted for differences in total body mass (i.e. W/kg). If total body mass is not corrected (when prescribing exercise based on metabolic heat production), Larger individuals working at the same metabolic rate as lighter people have a greater "heat sink" and therefore, requires more energy to raise their Tcore [12]. This mechanism is considered protective and often associated with a lower vulnerability to heat illness in larger individuals [12]. This further complicates the independent influence of body fat on thermoregulation as individuals with high body fat tend to be higher in body mass [12]. Moreover, individuals with high body fat tend to also have lower surface area to mass ratios, which translates to less surface area available for evaporation of sweat from the skin per unit body mass. In summary, it is difficult to ascertain whether differences in thermoregulatory response between Hi-BF

and Lo-BF are attributed to body fat when body surface area and body mass are also different between groups. Interestingly, the meta-analysis performed in the current study resulted in unclear ESs and non-significant findings. It is important to note that the number of interventions included were small, which makes it difficult to determine whether the findings were a lack of association or the result of type II error.

Dervis et al.[a] is arguably the most internally valid study to examine the effect of body fat on thermoregulation during exercise in the heat and should be modeled to examine the influence of independent factors on thermoregulation in the heat. This investigation matched their Hi-BF and Lo-BF groups in pairs for both total body mass and body surface area. Participants cycled for 60 min at 550W. To independently examine body fat, groups only differed in body fat percentage (difference of 21% between groups). Interestingly, they found that ΔTcore was significantly higher in the Hi-BF group compared to Lo-BF, with no differences in WBSL or Tsk. Dervis et al. (2015) reported that differences in ΔTcore with no corresponding thermoeffector response suggests differences in either physiological control (i.e. thermosensitivity or prolonged onset threshold) or conductivity of body tissues (i.e. low conductivity of fat tissue). By eliminating the confounding effects of total body mass, metabolic heat production, and body surface area through study design, there is stronger evidence to suggest that body fat may independently influence thermoregulatory responses. It is imperative that future work assessing the effect of body fat on thermoregulation during exercise in the heat control for these confounding variables in their study design. This will allow for a more comprehensive and precise evaluation of the true influence of adiposity as the result of a single study does not provide enough information to establish the state of existing knowledge.

The included studies present several other factors that may influence the relationship between body fat and thermoregulatory responses to exercise in heat. Dougherty et al. (2009) examined differences in thermoregulatory responses between obese and lean boys before and after they were heat acclimatized. Interestingly, Difference in ΔTcore between Day 6 (heat acclimatized) and Day 1(heat unacclimatized) was -0.23°C. and -0.05°C for lean and obese, respectively. In other words, although both lean and obese

groups experienced the physiological adaptations necessary to incur heat acclimatization, the obese boys acclimatized at a slower rate compared to the lean boys[21]. The participants in Dervis et al. (2016) were assumed to be unacclimatized to heat, but researchers were uncertain whether or not there were differences in heat acclimatization status between groups. Differences in heat acclimatization status can greatly affect thermoregulation and should be strongly considered in future investigations examining thermoregulatory responses to heat in Hi-BF vs. Lo-BF. Future investigations must also consider post-exercise Tcore. All Tcore data from included studies suggested that participants' Tcores did not exceed 38°C, which is below the threshold for exercise-induced hyperthermia(<38.5°C). Thermoeffector responses are considered a function of changes in body temperature and body temperature changes are heavily weighted by the influence of Tcore (opposed to Tsk)[40,41]. Therefore, as Tcore stayed below <38.5°C (i.e. hyperthermic zone) it is difficult to acertain whether differences in thermoeffector output would have occurred between Hi-BF and Lo-BF groups at higher Tcores. Lastly, the majority of the included investigations assessed steady state exercise. Eliciting steady state exercise may have diminished the influence of BF% on Tcore and WBSL, leading to no differences between groups [42].

Despite the many strengths of this systematic review and meta-analyses, there are limitations that must be addressed. First, the included studies only reflect children and adults under the age of 30. Future investigations should examine older adults as this aging is associated with differences in degree of adiposity and adipose tissue function as well as an enhanced risk of thermoregulatory dysfunction[43–45]. It should also be noted that although prepubertal children differ from adults in morphology and other physical factors, there does not appear to be maturational differences that influence children's ability to thermoregulate in the heat, compared to adults [46]. Additionally, the body fat groups were characterized by the authors of each individual study. Therefore, there was some overlap in BF% characterization between studies. Lastly, although a meta-analysis of studies controlling for metabolic heat production and body mass were performed, they were substantially underpowered. Aforementioned, underpowered analyses make it difficult to conclude whether the findings were the result of a lack of association or the result of type II error. As the literature currently stands, there are limited randomized controlled trials that

utilize the appropriate study designs to examine the independent influence of body fat on thermoregulation in the heat. To accurately examine the independent influence of body fat, randomized controlled trials using experimental approaches controlling for confounders of thermoregulatory response during exercise in the heat are warranted.

Conclusion

The included studies were heterogeneous and many did not control for confounding factors that affect thermoregulation in the heat such as metabolic heat production, body mass, body surface area, and heat acclimatization. Studies performing unbiased comparisons between Hi-BF and Lo-BF groups are recommended to further characterize the adverse effects of obesity on thermoregulation in the heat [15,36,38]. Future investigations should also examine differences in Hi-BF and Lo-BF groups who were hyperthermic. It is possible that differences in thermoeffector responses between groups are a function of changes in thermoeffector onset threshold or thermosensitivity, that may only be present after higher Tcores.

Practical Implications

- There are current methodological concerns in research studies surrounding the influence of body fat on thermoregulation during exercise in the heat
- Future investigations need to make unbiased comparisons (i.e. control for confounding factors such as metabolic heat production, body mass, heat acclimatization, body surface area, etc.) between Hi-BF and Lo-BF to examine the independent influence of body fat on thermoregulation during exercise in the heat.

Acknowledgements

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Appendix A. Supplemental Data

Supplemental data associated with this article can be found in the online version.

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Table 1. Descriptive Characteristics and Outcomes of Included Interventions (K=18).

Studies	(n)				(%)	(%)	Groups	Modality		(min)		(ES, 95% CI)		WBSL (ES, 95% CI)
Adams et al. (2015)[a] [18]	10	10	Sedentary Females	Hi-BF: 26 Lo-BF: 27	43.5	26.3	17.2	Cycling	300W Fixed MHP	60	40°C T_{amb}; 30% RH	0.85 [-0.03, 1.73] **Unclear**	0.15 [-0.73,1.03] **Unclear**	-0.85 [-1.77, 0.06] **Unclear**
Adams et al. (2015)[b] [18]	10	10	Sedentary Females	Hi-BF: 26 Lo-BF: 27	43.5	26.3	17.2	Cycling	175W/m^2 Fixed MHP per BSA	60	40 °C T_{amb}; 30% RH	0.76 [-0.11, 1 .64] **Unclear**	1.10 [0.08, 1.94] **Very Large**	0.00 [-0.88, 0.88] **Unclear**
Cramer & Jay (2014)[a] [14]	8	8	Physically Active Males, Groups matched	Hi-BF: 24 Lo-BF: 24	22.0	12.5	9.5	Cycling	9.0W/kg of BM Fixed MHP per BM	60	34.7°C T_{amb}; 34.1% RH	1.31 [0.28, 2.34] **Very Large**	-0.07 [-1.05, 0.91] **Unclear**	0.19 [-1.01, 1.39] **Unclear**
Cramer & Jay (2014)[b] [14]	8	8	Physically Active Males, Groups matched	Hi-BF: 24 Lo-BF: 24	22.0	12.5	9.5	Cycling	9.0W/kg of BM Fixed MHP per BM	60	25°C T_{amb}; 36.8% RH	1.32 [0.29, 2.39] **Very Large**	-0.49 [-0.98, -0.00] **Moderate**	
Cramer & Jay (2014)[c] [14]	8	8	Physically Active Males, Groups matched	Hi-BF: 24 Lo-BF: 24	22.0	12.5	9.5	Cycling	500W	60	25°C T_{amb}; 36.8% RH	-0.33 [-1.27, 0.60] **Unclear**	-1.57 [-2.08, -1.06] **Very Large**	
Cramer & Jay (2014)[d] [14]	8	8	Physically Active Males, Groups matched	Hi-BF: 24 Lo-BF: 24	22.0	12.5	9.5	Cycling	6.5W/kg of BM Fixed MHP per BM	60	25°C T_{amb}; 36.8% RH	0.74 [-0.23, 1.70] **Unclear**	-0.32 [-1.07, 0.43] **Unclear**	
Cramer & Jay (2014)[e] [14]	8	8	Physically Active Males, Groups matched	Hi-BF: 24 Lo-BF: 24	22.0	12.5	9.5	Cycling	600W	60	25°C T_{amb}; 36.8% RH	0.19 [-0.74, 1.12] **Unclear**	-1.89 [-2.39, -1.39] **Very Large**	
Dervis et al. (2016)[a] [15]	9	9	Physically Active Males; matched in pairs for body mass, body surface area	Hi-BF: 25 Lo-BF: 24	32.0	10.8	21.2	Cycling	550W Fixed MHP	60	28°C T_{amb}; 26% RH	0.51 [-0.38, 1.41] **Unclear**	4.97 [3.10, 6.86] **Very Large**	0.28 [-0.65, 1.21] **Unclear**
Dervis et al. (2016)[b] [15]	9	9	Physically Active Males; matched in pairs for body mass, body surface area	Hi-BF: 25 Lo-BF: 24	32.0	10.8	21.2	Cycling	7.5W/kg of LBM Fixed MHP per LBM	60	28°C T_{amb}; 26% RH	-0.25 [1.14, 0.63] **Unclear**	-2.53 [-3.77, -1.29] **Very Large**	0.79 [-0.17, 1.75] **Unclear**
Dougherty et al. (2009)a	7	7	Physically Active Heat-unacclimatized Boys	Hi-BF: 11 Lo-BF:11	33	18	15	Cycling and Treadmill Walking	30% VO2max	60	38°C T_{amb}; 50% RH	-	-1.78 [-3.02, -0.55] **Very Large**	1.09 [-0.03, 2.20] **Unclear**
Dougherty et al. (2009)b	7	7	Physically Active Heat Acclimatized Boys	Hi-BF: 11 Lo-BF: 11	33	18	15	Cycling and Treadmill Walking	30% VO2max	60	38°C T_{amb}; 50% RH	-	2.86 [1.38, 4.36] **Very Large**	1.60 [0.40, 2.81] **Very Large**
Leites et al. (2013)	13	14	Prepubertal girls	Hi-BF: 9 Lo-BF: 9	41.3	16.6	24.7	Cycling	50-55% VO2peak	30	~35°C T_{amb}; 40% RH	-	-1.26 [-2.08, -0.43] **Very Large**	0.90 [0.11, 1.69] **Large**
Limbaugh et al. (2013) [19]	8	9	Recreationally active, unacclimatized Males; matched for fitness	Hi-BF: 26 Lo-BF: 25	23.6	11.3	12.3	Cycling	66 W Resistance Absolute Workload	60	30°C T_{amb}; 30% RH	0.30 [-0.61, 1.21] **Unclear**	0.02 [-0.96, 1.0] **Unclear**	-
Rodrigues et al. (2020)a	12	7	Fit Pubescent Girls	Hi-BF: 13 Lo-BF: 13	41.0	25.1	15.9	Cycling	~5.4 W/kg Fixed MHP per BM	50	36°C T_{amb}; 40% RH	-	-	0.00 [-0.94,0.95] **Unclear**
Rodrigues et al. (2020)b	9	8	Unfit Pubescent Girls	Hi-BF: 13 Lo-BF: 13	39.9	23.3	16.6	Cycling	~5.4 W/kg Fixed MHP per BM	50	36°C T_{amb}; 40% RH	-	-	0.23 [-0.70, 1.16] **Unclear**
Tucker et al. (2017) [16]	9	11	Healthy Males	Hi-BF: 26 Lo-BF: 24	30.2	13.6	16.6	Cycling	6 W/kg; Fixed MHP per BM	60	40.3°C T_{amb}; 32.5% RH	-	1.30 [0.28, 2.32] **Very Large**	-1.72 [-2.81,- 0.64] **Very Large**
Sehl et al. (2011)	17	16	Physically Active Pubescent Boys	Hi-BF: 13 Lo-BF: 13	68.6	12.1	56.5	Cycling	50-55% of VO2peak	30	35°C T_{amb}; 45% RH	-	-1.58 [-2.35, -	-

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													0.79] **Very Large**	
Vroman et al. (1983)	5	5	Healthy Sedentary Males	Hi-BF: 19-27 Lo-BF: 19-27	32.3	16.8	15.5	Cycling	50% VO_2max (one exercise arm)	25-30min	35°C T_{amb}	-0.42 [-1.55, 0.72] **Unclear**	0.04 [-1.20, 1.25] **Unclear**	-

Note: [a,b,c] sub-studies included in meta-analysis which were separated by exercise trial. Hi-BF, high body fat group; lo-BF, low body fat group, MHP, metabolic heat production; diff, difference; W, watts; kg, kilogram; hr, hour; BSA, body surface area; BM, total body mass; m^2, meters squared; LBM, lean body mass; T_{amb}, ambient temperature; RH, relative humidity; Tcore, core temperature; Tsk, mean skin temperature; SR, whole body sweat rate; %, percentage; n, sample size; °C, degrees Celsius.

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Figure Legends

Figure 1. PRISMA flowchart detailing systematic search and selection process of included studies

Figure 2. Exercise intensity prescription methods used in included interventions. Note: W, watts; W/BSA, watts per unit of body surface area; W/Lean Kg, watts per kilogram of lean body mass; W/kg, watts per kilogram of total body mass.

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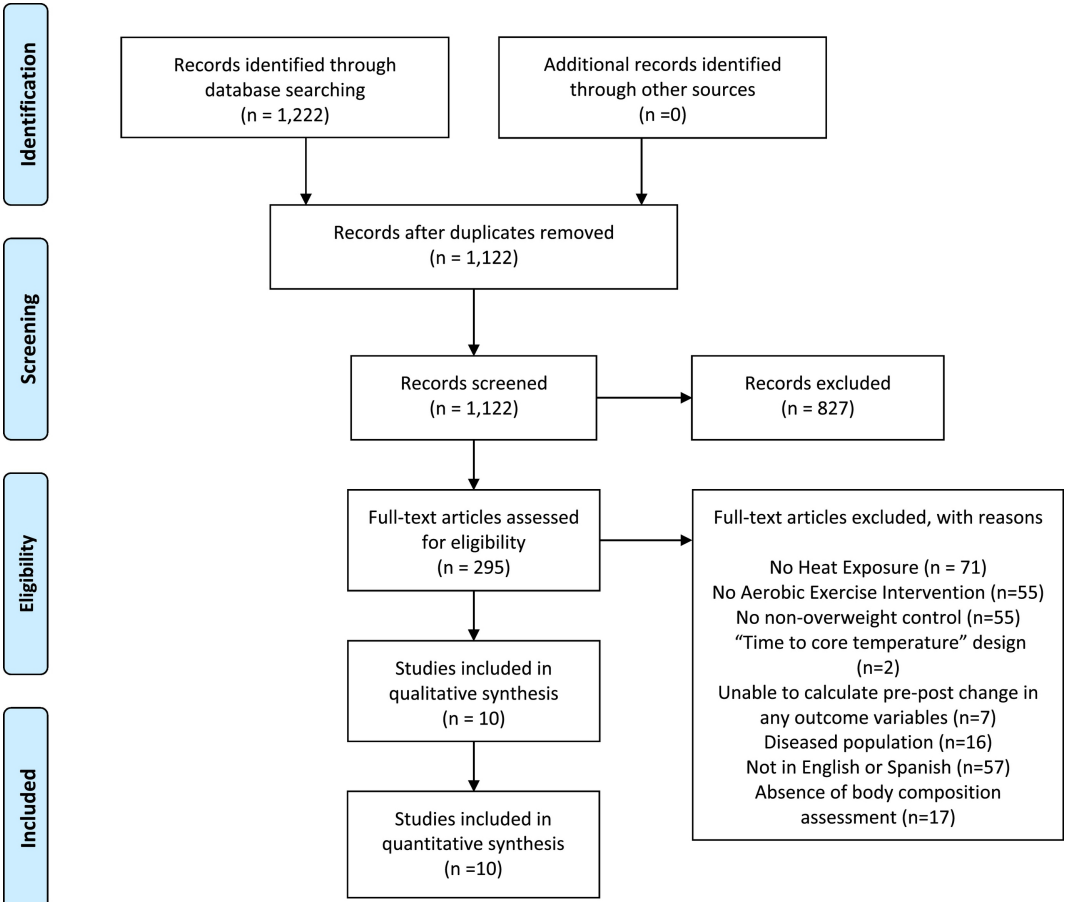
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Screening
Eligibility
Included
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Additional records identified through other sources (n =0)
Records after duplicates removed (n = 1,122)
Records screened (n = 1,122)
Records excluded (n = 827)
Full-text articles assessed for eligibility (n = 295)
Full-text articles excluded, with reasons
No Heat Exposure (n = 71)
No Aerobic Exercise Intervention (n=55)
No non-overweight control (n=55)
"Time to core temperature" design (n=2)
Unable to calculate pre-post change in any outcome variables (n=7)
Diseased population (n=16)
Not in English or Spanish (n=57)
Absence of body composition assessment (n=17)
Studies included in qualitative synthesis (n = 10)
Studies included in quantitative synthesis (n =10)

Figure 1

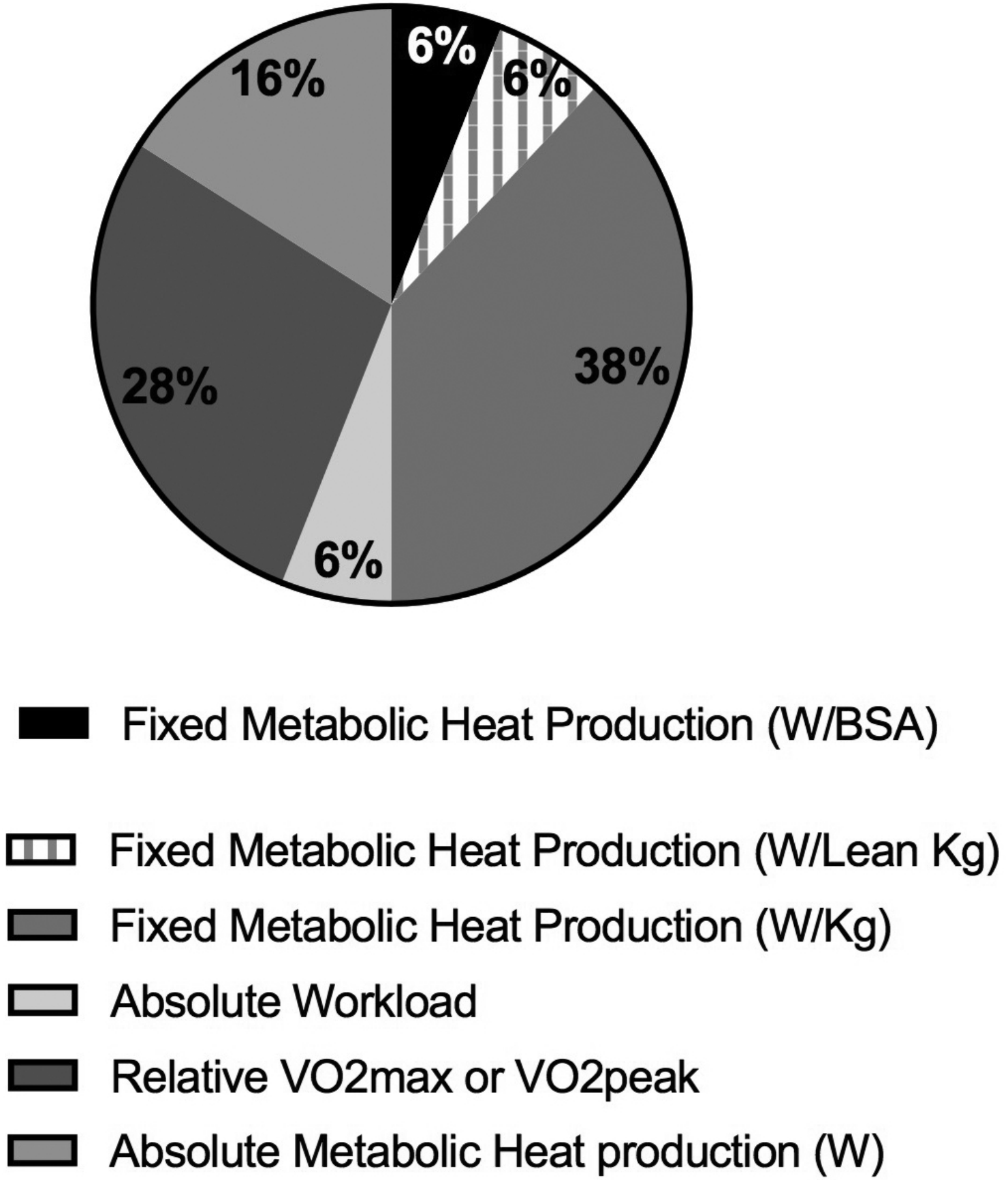
6%
6%
16%
38%
28%
6%
Fixed Metabolic Heat Production (W/BSA)
Fixed Metabolic Heat Production (W/Lean Kg)
Fixed Metabolic Heat Production (W/Kg)
Absolute Workload
Relative VO2max or VO2peak
Absolute Metabolic Heat production (W)

Figure 2